THE CASE FOR MULTIPLE MEMORY OR KNOWLEDGE STORES

In more advanced systems, we'll look at agents employing multiple memory and knowledge stores relevant to their workflow. For example, an agent could employ individual memory stores as part of its conversations with individual users, perhaps including the ability to share different groups of memory with different groups of individuals. Memory and knowledge retrieval are cornerstones of agentic systems, and we can now summarize what we covered and review some learning exercises in the next section.

8.8 Exercises

Use the following exercises to improve your knowledge of the material:

- *Exercise 1* —Load and Split a Different Document (Intermediate)

Objective —Understand the effect of document splitting on retrieval efficiency by using LangChain.

Tasks:

  - Select a different document (e.g., a news article, a scientific paper, or a short story).
  - Use LangChain to load and split the document into chunks.
  - Analyze how the document is split into chunks and how it affects the retrieval process.
- *Exercise 2* —Experiment with Semantic Search (Intermediate)

Objective —Compare the effectiveness of various vectorization techniques by performing semantic searches.

Tasks:

  - Choose a set of documents for semantic search.
  - Use a vectorization method such as Word2Vec or BERT embeddings instead of TF–IDF.